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Course Name – Data Analytics With GenAI

Subject – PowerBI

Topic – PowerBI Visuals Assignment

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PowerBI File : File Link

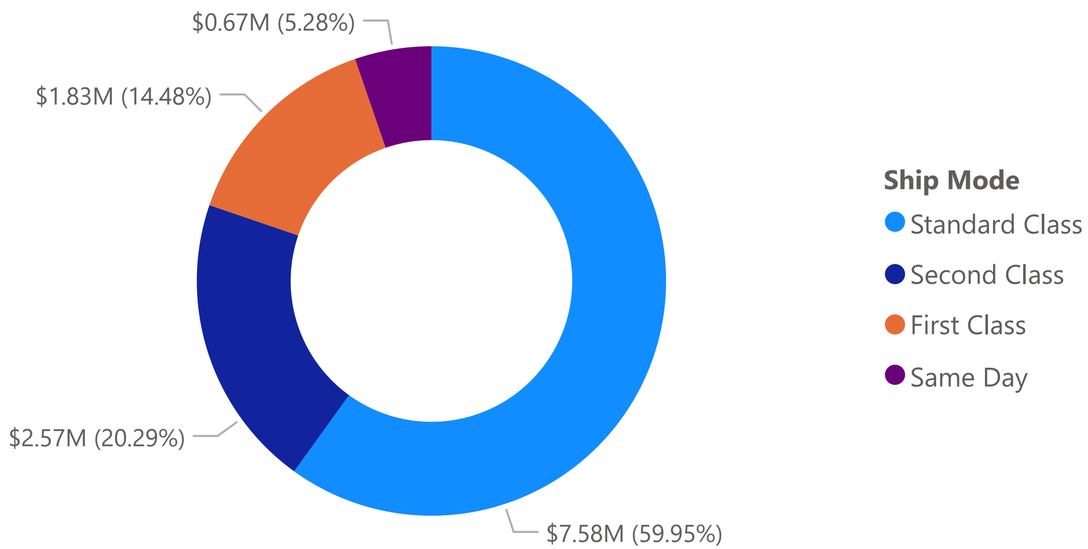
## Question 1 : Create a Card Visual to show Total Sales.

$12.64M

Sum of Sales

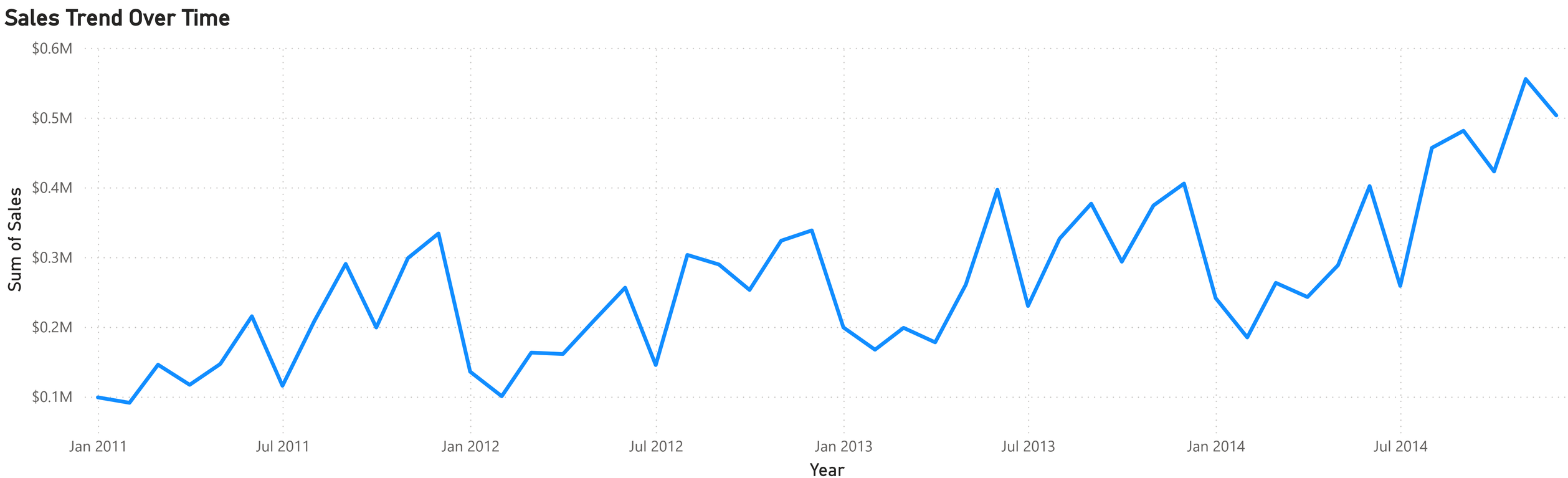
# Question 2 : Create a Donut Chart showing Total Sales by Ship Mode and which ship mode generates the highest sales?

## Total Sales by Ship Mode




## Standard Class generates the highest sales

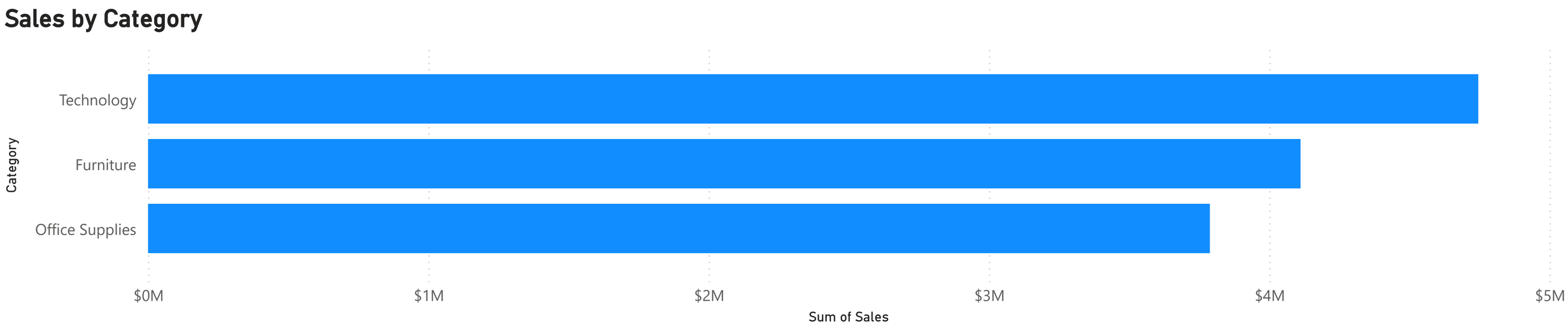
## Question 3 : Create a Line Chart showing the Sales Trend over time (by Month and Year). Is there any noticeable seasonal pattern in sales?




Yes! A clear pattern is visible:

| | |
|---|---|
| **Peak Sales** | **November & December** every year (holiday season) |
| **Low Sales** | **January & February** (post-holiday drop) |
| **Mid-year rise** | Sales pick up again around **August–September** |
| **Year-on-year** | Overall sales are **growing each year** |

# Question 4 : Create a Bar Chart to compare Sales by Category




| Rank Sales | Category | Approx. |
|---|---|---|
| 1st | **Technology** | ~$4.7M |
| 2nd | **Furniture** | ~$4.1M |

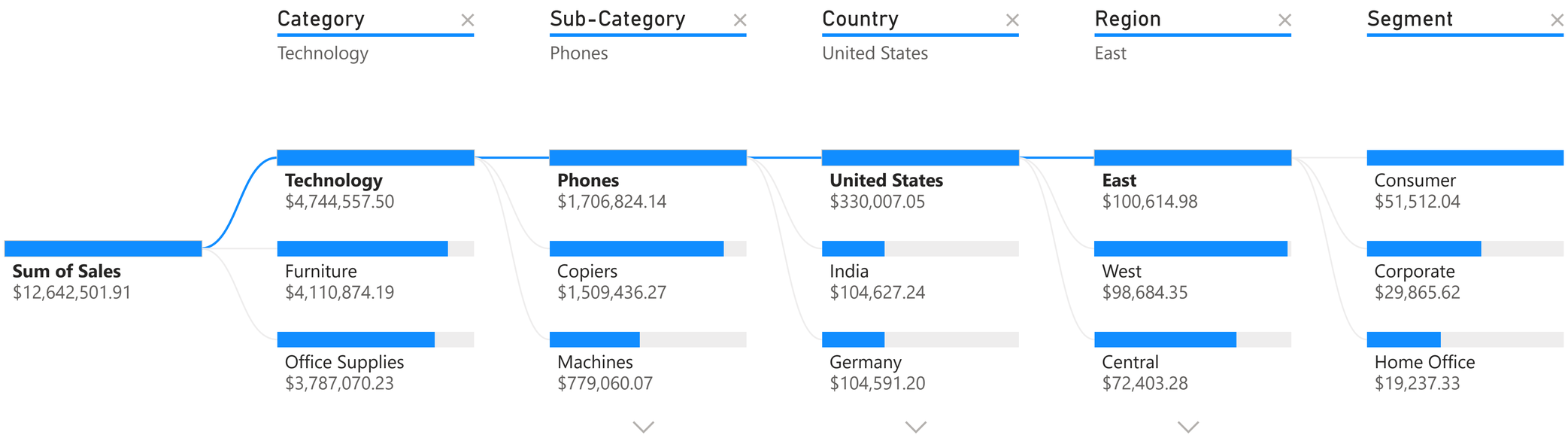
# Question 5 : Create a Matrix Visual to display Sales by Month and Category and identify the month with the highest sales.

## Sales by Month and Category

| Month | Furniture | Office Supplies | Technology | Total |
|---|---|---|---|---|
| February | $185,359.81 | $153,504.36 | $204,875.22 | **$543,739.38** |
| January | $228,566.69 | $222,235.38 | $224,331.61 | **$675,133.67** |
| April | $211,210.05 | $227,602.25 | $259,748.91 | **$698,561.21** |
| July | $227,108.22 | $234,934.07 | $287,339.55 | **$749,381.84** |
| March | $254,891.14 | $227,700.56 | $287,909.24 | **$770,500.94** |
| May | $280,688.62 | $279,817.91 | $343,505.81 | **$904,012.34** |
| October | $382,255.36 | $300,781.71 | $485,147.32 | **$1,168,184.39** |
| June | $411,266.10 | $372,781.29 | $485,669.36 | **$1,269,716.75** |
| August | $393,218.25 | $402,527.97 | $498,086.95 | **$1,293,833.17** |
| September | $494,672.03 | $426,931.13 | $515,776.95 | **$1,437,380.11** |
| November | $502,717.07 | $468,904.46 | $579,655.81 | **$1,551,277.34** |
| December | $538,920.85 | $469,349.15 | $572,510.77 | **$1,580,780.77** |
| **Total** | **$4,110,874.19** | **$3,787,070.23** | **$4,744,557.50** | **$12,642,501.91** |

| Rank | Month |
|---|---|
| Approx. 1st ~$1.8M | **December** |

# Question 6 : Create a Decomposition Tree for Sales Breakdown.



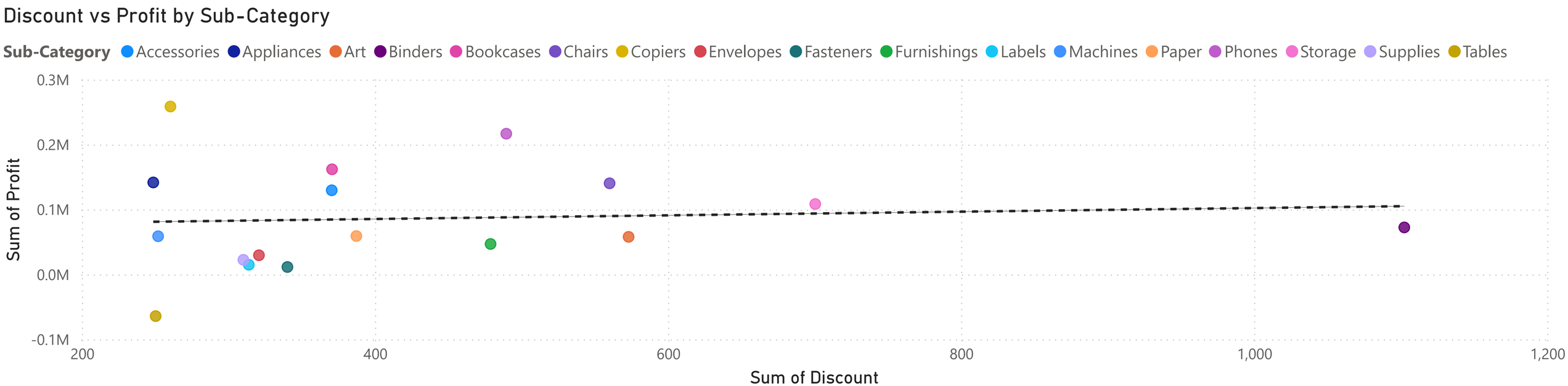

Overall Insights:

. **Technology** → **Phones** → **USA** → **East** → **Consumer** is the most profitable path

. **US Market dominates** Phone sales by a large margin over India & Germany

. **Consumer segment** drives the most sales in the East region

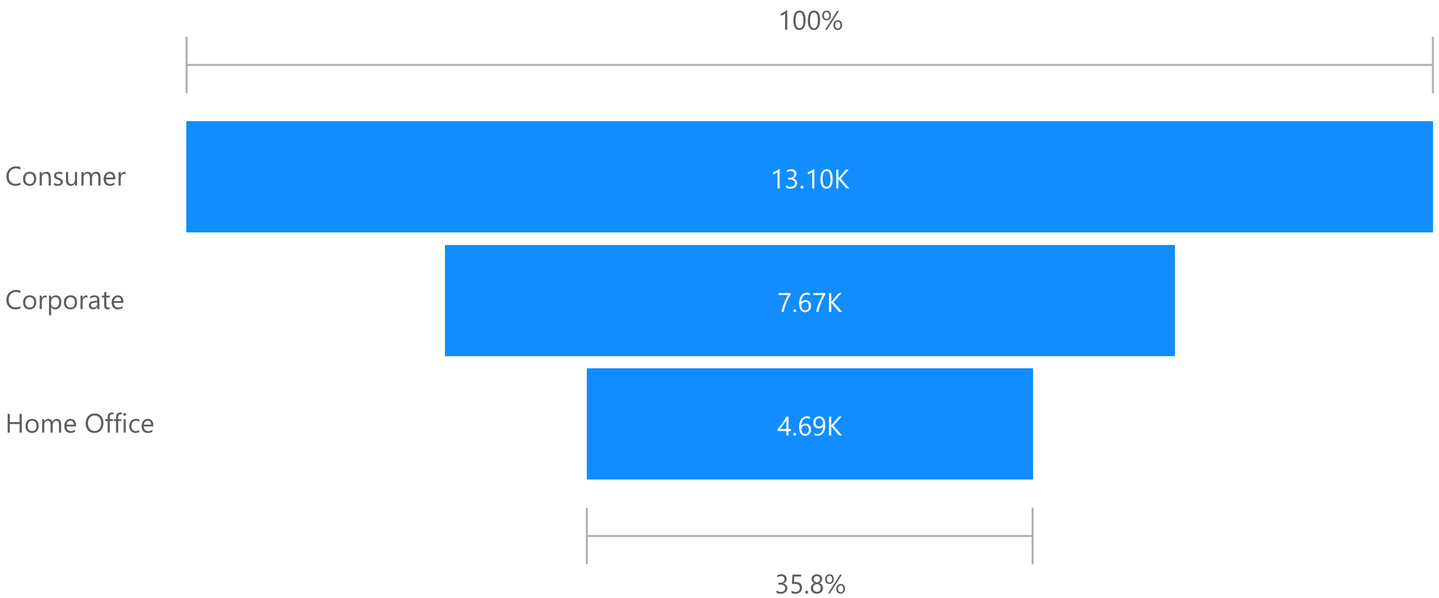
# Question 7 : Create a Scatter Chart to compare Discount vs. Profit by Sub-Category. Are there any outliers where high discounts resulted in negative profit?




## Yes, there ARE outliers where high discounts caused negative profit

# Question 8 : Use a Funnel Chart to show the number of orders by Customer Segment.

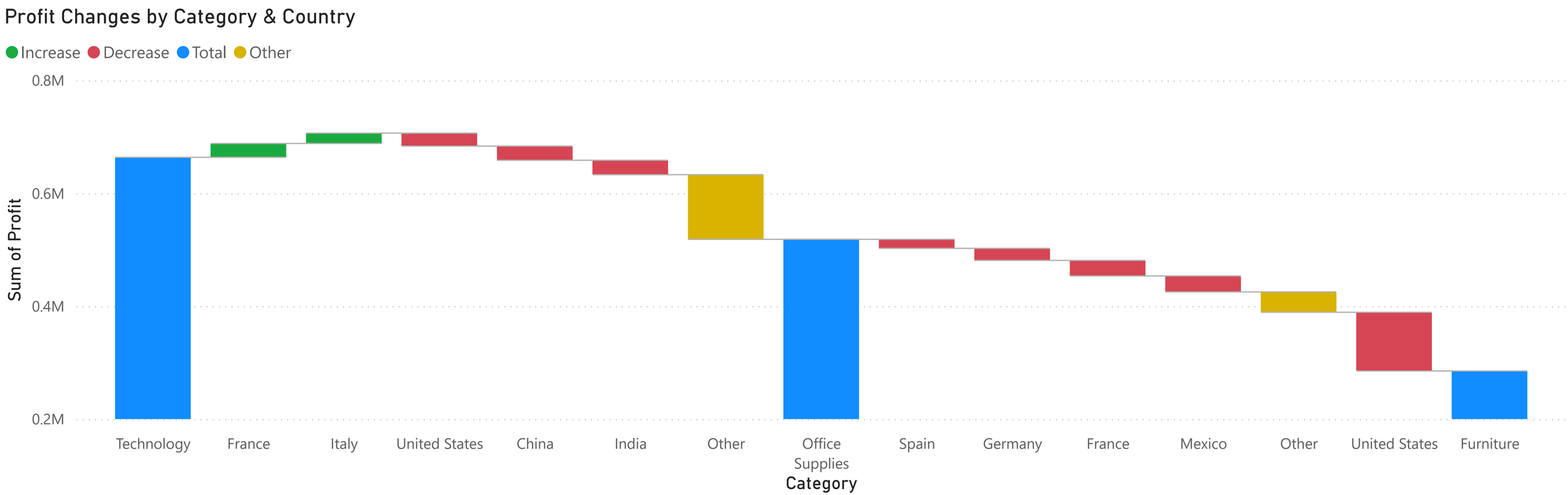
Number of Orders by Customer Segment




Key Insights:

- **Consumer segment dominates** with more than half of all orders
- **Corporate** is second — fewer but likely higher value orders
- **Home Office** has the least orders but shouldn't be ignored
- The funnel shape visually shows the **drop-off** between segments

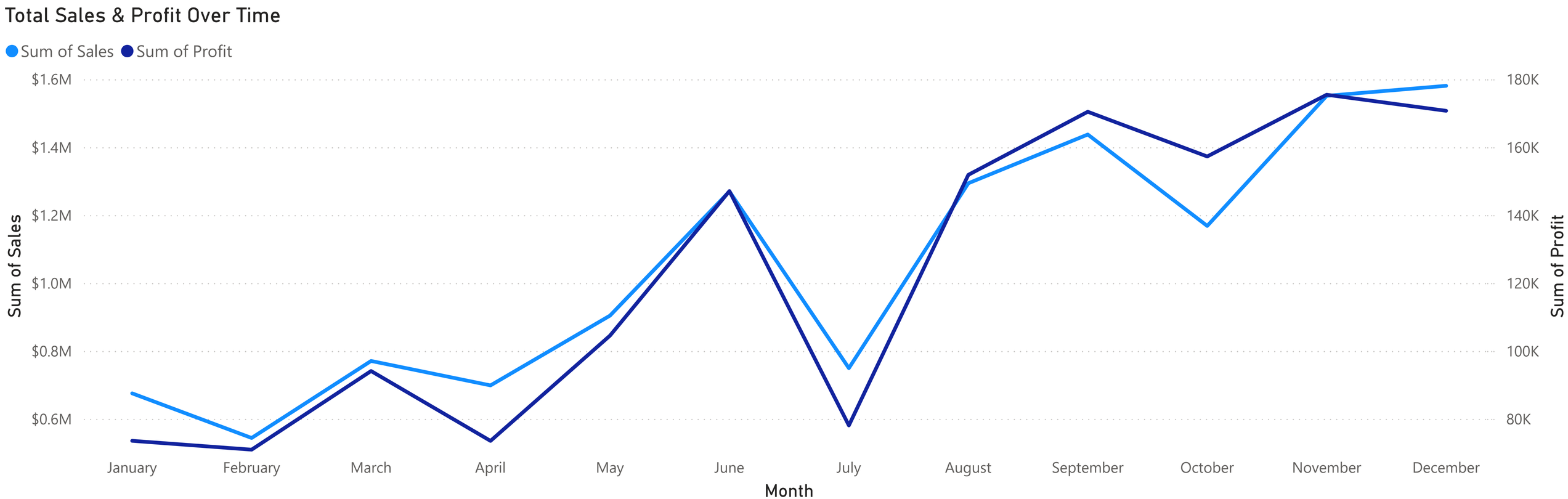
# Question 9 : Use a Waterfall Chart to analyze how Profit changes by Category within the countries.




Key Insights from chart:

- **Technology is the most profitable** category (~$0.66M)
- **Furniture has the lowest profit** (~$0.27M) after country breakdowns
- **United States under Furniture** shows a significant profit drop
- **France & Italy** are strong contributors to Technology profit

# Question 10 : Create a Dual-Axis Line Chart to show Total Sales and Profit Over Time (by Month).




Key Insights from Chart:

**Low Months:**

- **January & February** → Both Sales & Profit are at their lowest
- **July** → Surprising sharp dip! Both lines drop suddenly

**Peak Months:**

- **June** → Unexpected mid-year spike in Profit (dark line jumps high!)
- **November & December** → Both Sales & Profit reach their highest